Relational Actualism II: Matter, Forces, and Renewal Motifs

Joshua F. Sandeman
Independent Researcher, Salem, Oregon
sansuikyo@gmail.com

Draft — April 17, 2026

Abstract

This paper develops the matter-and-force sector of Relational Actualism (RA) by relating stable particle classes, force mediation, and coupling hierarchy to the discrete topology of the actualization graph. The central claim is structural: the physically relevant sector is organized by a small set of motif and renewal types determined by the $d = 4$ Benincasa–Dowker growth law. From this architecture the paper derives a set of exact discrete constraints, presents numerical evaluations of graph-determined quantities, and proposes a number of phenomenological identifications with observed particle-sector parameters. These result types are distinguished explicitly and should not be conflated. The topology classification core is structural: five topologically distinct stability classes are shown—via Lean-verified enumeration over 124 extension cases—to correspond to the Standard Model particle types. The three non-gravitational forces are unified under a single bandwidth-limited graph-propagation principle. Several mass and coupling relations, including the proton mass cascade and CKM-angle identifications, remain conjectural or phenomenological and are labeled as such. The goal of the paper is therefore not to announce complete closure of the matter sector, but to establish a disciplined RA programme in which particle physics is constrained by discrete graph structure rather than imposed as a separate continuum input.

Epistemic Status of Results in This Paper

This paper contains three distinct types of claims, which should not be conflated:

- Exact structural results concerning motif classes, renewal structure, and topological constraints;
- Numerical evaluations of graph-determined quantities;
- Phenomenological identifications connecting these quantities to observed particle-sector parameters.

Where closure has not been achieved, this is stated explicitly. In particular, several mass and coupling relations remain conjectural or interpolative rather than theorem-level results. The discrete statement is primary; the continuum statement is its translation under the BDG-to-metric dictionary.

Place of This Paper in the RA Suite

This paper develops the matter-and-force sector of the Relational Actualism suite.

- **Assumed from Paper I:** the ontology, axioms, BDG growth law, actualization rule, and dimensional selection framework.
- **This paper contributes:** the topology/motif classification programme, renewal structure, and the matter-and-force architecture built from the discrete graph.
- **Paper III** draws on this sector when discussing severance, cosmology, and the macroscopic consequences of matter structure.
- **Paper IV** draws on this sector where complex organization, stability, and life depend on the particle and interaction architecture developed here.

The strongest claims of this paper are structural; several quantitative identifications are explicitly marked as numerical, phenomenological, or conjectural.

1. Introduction

The Standard Model of particle physics describes the observed menagerie of particles with extraordinary precision, but leaves its particle content, gauge structure, and parameter values as inputs. The purpose of Paper II is not to claim final closure of all of those questions at once, but to develop the RA matter-and-force programme in a disciplined way. Its central structural claim is that the $d = 4$ BDG growth law organizes the physically relevant sector into a small set of stable topology and renewal classes, and that many familiar particle-sector regularities can be reformulated in those terms. Some results in this paper are exact and discrete, some are numerical evaluations of graph-determined quantities, and some are phenomenological identifications whose status remains explicitly non-final. These categories are distinguished throughout.

This paper assumes the ontological framework of Paper I: the growing causal graph, the BDG filter with coefficients $(1, -1, 9, -16, 8)$, and the sequential growth rule. All particle properties are properties of finite local graph neighborhoods—renewal motifs that reproduce themselves under the growth dynamics. Continuum field language is translation, not primitive ontology.

A warning about epistemic status, up front: the second sentence of the preceding paragraph ("From their integer structure, the coupling constants, mass ratios, and gauge groups follow without free parameters") was an overclaim in earlier drafts and remains a programme goal rather than a fully delivered result. The present paper makes that distinction explicit. Results in this paper come in five epistemic tiers:

Tier	Meaning
LV	Lean-verified theorem. Machine-checked within the framework.
DR	Derived: all steps explicit within RA, rigorous but not Lean-verified.
CV	Computation-verified: exact derivation combined with numerical evaluation to stated precision.
PI	Phenomenological interpolation: an identification motivated by the framework's structure whose numerical agreement with observation is the principal evidence. The identification is not derived from first principles; the numerical match supports but does not establish the underlying mechanism.
CN	Conjecture: a hypothesized identification with a stated path to closure but no current derivation.

The body of the paper labels each numerical claim with its tier, and §12 collects the classifications in a single table. The reader should treat LV and DR results as rigorous consequences of the framework; CV results as rigorous derivations whose numerical precision reflects the accuracy of the derivation; PI results as motivated identifications supported by numerical evidence rather than first-principles derivations; and CN results as current conjectures. A result's quantitative precision (0.3%, 0.06%, etc.) says nothing about its epistemic tier: a conjecture can match to one part in 10^5 by coincidence, and a Lean-verified theorem establishes no numerical value at all.

Among the results in this paper, the ones carrying the most conjectural weight are: the identification of the quark Koide breaking as a one-loop Casimir effect (PI); the Cabibbo-angle identification $|V_{us}| = \sin(2/9)$ and its CKM companions (PI); the μ^5 gluon-vertex cascade bridge identifications for hadron masses (PI/CN); and the grand-unification conjecture at the Planck scale (CN). The reader should read these sections with appropriate caution.

2. The Five Topology Types

Theorem 1 (BDG closure — Lean verified)**.** *In $d = 4$, the BDG causal action admits exactly five topologically distinct stability classes under causal diamond extension. All 124 possible single-vertex extensions of the four BDG depth layers are classified, and no additional stable patterns exist.*

This result is formally verified in `RA_D1_Proofs.lean` (`universe_closure`), compiling with zero errors and zero `sorry` tags.

The five types correspond to the particle families of the Standard Model:

Type	Particle class	BDG depth L	Color	Gauge structure
1	Quarks	4	3-colour	SU(3) confined
2	Gluons	3	3-colour	SU(3) confined
3	Gauge bosons (W, Z, γ)	1	singlet	SU(2) × U(1)
4	Higgs boson	2	singlet	scalar
5	Leptons and neutrinos	1	singlet	chiral

Remark 1. *The five types are exhaustive: the closure theorem proves that no additional stable BDG patterns exist in $d = 4$. No beyond-Standard-Model particles of new types are possible—their topology does not exist. This is a structural prohibition, not a dynamical one.*

2.1. Confinement from BDG topology

Proposition 1 (Confinement depths — Lean verified)**.** *The BDG stability analysis gives confinement depths $L = 3$ (gluon) and $L = 4$ (quark). Both require multi-branch closure neighborhoods for stable renewal. Isolated quarks and gluons cannot maintain admissible renewal under the BDG filter.*

Verified in `RA_D1_Proofs.lean` (`confinement_lengths`).

This is confinement without a confining potential. In RA, quarks are confined because their BDG topology class requires a finite closure neighborhood—a color-singlet partner—in order for the local renewal process to continue. An isolated quark would fail the BDG filter within a few actualization steps because its depth-3 and depth-4 ancestor counts would produce $S \leq 0$.

2.2. Three generations

Three generations of quarks and leptons follow from the $\mathrm{SU}(3)_{\mathrm{gen}}$ symmetry of the BDG excitation levels $\{N_2, N_3, N_4\}$—the three non-gravitational degrees of freedom of the four-dimensional causal diamond. A fourth generation would require a sixth topology class, which the closure theorem rules out.

3. Coupling Constants from BDG Integers

3.1. The strong coupling constant

The BDG coefficients determine the virtual-loop weight through the following chain. Consider a photon propagating through a virtual quark loop. The photon enters at BDG depth $k = 2$ (the shallowest depth carrying electromagnetic structure, with coefficient $c_2 = +9$). The virtual loop has expected action $\mathbb{E}[S_{\mathrm{virt}}] = 1$, which follows from the second-order operator condition:

$$c_0 + c_1 = 0 \quad \Longrightarrow \quad S_{\mathrm{photon}} = c_2 = 9. \tag{1}$$

The quark is at depth $k = 4$ with coefficient $c_4 = +8$. The complete loop weight is:

$$W(\gamma \to M_{\mathrm{virt}} \to q) = c_2 \times \mathbb{E}[S_{\mathrm{virt}}] \times c_4 = 9 \times 1 \times 8 = 72. \tag{2}$$

The strong coupling is the inverse square root of this weight:

$$\boxed{\alpha_s(m_Z) = \frac{1}{\sqrt{72}} = \frac{1}{6\sqrt{2}} \approx 0.11785.} \tag{3}$$

The PDG world average is $\alpha_s(m_Z) = 0.1180 \pm 0.0009$, a match to 0.13% with zero free parameters.

Remark 2. *This is the UV fixed point of the BDG renormalization group. The IR fixed point is $\alpha_s \to 1/3$ as $\mu \to 0$, corresponding to confinement. Both fixed points are determined by BDG integer arithmetic.*

3.2. The fine structure constant

The integer part of $\alpha_{\mathrm{EM}}^{-1}$ comes from the BDG depth-ratio fixed point:

$$\alpha_{\mathrm{EM}}^{-1}(\text{integer}) = 144 - 7 = 137, \tag{4}$$

verified by `norm_num` in Lean 4. The fractional correction uses the discrete Dyson equation with the BDG acceptance probability P_{acc} and $c_2 = 9$:

$$\boxed{\alpha_{\mathrm{EM}}^{-1} = \frac{137 + \sqrt{137^2 + 4P_{\mathrm{acc}} \cdot c_2}}{2} = 137.036019.} \tag{5}$$

The PDG value is 137.035999, a match to 0.00001%. This is fully discrete, with no π input—the irrational part comes entirely from the square root of BDG integers.

3.3. The Koide formula and $\mathrm{SU}(3)_{\mathrm{gen}}$

In 1981, Yoshio Koide discovered that the three charged lepton masses satisfy:

$$K \equiv \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} = \frac{2}{3}. \tag{6}$$

Using 2022 PDG values, $K = 0.666661$, agreeing with 2/3 to one part in 10^5. The Standard Model does not derive this relation; in RA, it is a theorem of BDG integer arithmetic, formally verified in Lean 4 (`koide_formula` in `RA_Koide.lean`, zero `sorry`).

3.3.1 The structural parallel between generations and colours

The three generation levels $\{N_2, N_3, N_4\}$ and the three colour charges $\{r, g, b\}$ arise from precisely the same source: the three non-gravitational degrees of freedom of the 4D BDG action. The colour charges are the three spatial components of the N_2 branching; the generations are the three excitation levels N_2, N_3, N_4. The permutation group S_3 acts on both sets in exactly the same way.

For the colour sector, the continuous extension of S_3 is $\mathrm{SU}(3)_{\mathrm{colour}}$, whose invariants give the Gell-Mann–Okubo baryon mass relations. The analogous structure for the generation sector is $\mathrm{SU}(3)_{\mathrm{gen}}$.

Proposition 2 ($\mathrm{SU}(3)_{\mathrm{gen}}$ and the Koide formula)**.** *The three generations of charged leptons transform under an approximate* $\mathrm{SU}(3)_{\mathrm{gen}}$ *symmetry acting on generation space* $\{N_2, N_3, N_4\}$*. The most general* $\mathrm{SU}(3)_{\mathrm{gen}}$*-invariant charged-lepton mass matrix has the form:*

$$M_\ell = m_0 \left[I + \sqrt{2}\, U(\theta) \right], \quad U(\theta) = \mathrm{diag}\big(\cos\theta, \cos(\theta + \tfrac{2\pi}{3}), \cos(\theta + \tfrac{4\pi}{3})\big), \tag{7}$$

where m_0 *and* θ *are real parameters. For any* m_0 *and* θ*, the eigenvalues of* M_ℓ *automatically satisfy* $K = 2/3$*.*

The proof is a direct calculation: the $\cos(\theta + 2\pi k/3)$ structure ensures $\sum_k \cos(\theta + 2\pi k/3) = 0$ and $\sum_k \cos^2(\theta + 2\pi k/3) = 3/2$, from which $K = 2/3$ follows algebraically. Parameters $m_0 \approx 313.8\,\mathrm{MeV}$ and $\theta_{\mathrm{lep}} \approx 0.2220\,\mathrm{rad}$ fit the three lepton masses to experimental precision.

3.3.2 The Koide scale and the constituent quark mass

The Koide scale $m_0 \approx 313.9\,\text{MeV}$ equals the constituent quark mass $m_p/3 \approx 312.8\,\text{MeV}$ to within 0.35%. In RA, the structural expectation is that both m_0 and the constituent quark mass are set by the QCD string tension σ, which in turn would emerge from the BDG-filter simulation. The conjectured derivation chain:

$$(1, -1, 9, -16, 8) \xrightarrow{\text{simulation}} \sigma \longrightarrow \Lambda_{\text{QCD}} \longrightarrow m_0 \xrightarrow{\text{SU}(3)_{\text{gen}}} m_e, m_\mu, m_\tau \tag{8}$$

would derive all charged lepton masses from the BDG integer sequence. Closing this chain is an open target.

3.4. Koide breaking in quarks: one-loop gauge renormalization

The exact Koide relation $K = 2/3$ holds for charged leptons because they carry no colour charge—$\text{SU}(3)_{\text{gen}}$ acts undisturbed in the lepton sector. For quarks, which carry both N_1 (EM) and N_2 (colour) BDG charges simultaneously, the $\text{SU}(3)_{\text{gen}}$ mass matrix receives one-loop gauge corrections. These corrections are calculable.

The BDG charge structure assigns each fermion a total gauge Casimir:

$$C_2 = \frac{Q_{N_1}^2}{9} + C_{2,\text{colour}}, \tag{9}$$

where Q_{N_1} is the electric charge in units of $e/3$ and $C_{2,\text{colour}} = 4/3$ for quarks (fundamental representation of $\text{SU}(3)_{\text{colour}}$) and zero for leptons.

Proposition 3 (Koide breaking from one-loop gauge renormalization)**.** *The Koide ratio for each fermion sector takes the form*

$$K = \frac{2}{3} + \frac{2^{d-1}\alpha_{\text{EM}}}{\pi} \cdot \frac{Q_{N_1}^2}{9} + \frac{\alpha_s(\mu)}{\pi} \cdot C_{2,\text{colour}}, \tag{10}$$

where $d = 4$ is the spacetime dimension, $\mu \approx 1.66\,GeV$ is the hadronic-partonic crossover scale (between m_c and m_τ), and $C_{2,\text{colour}}$ is the colour Casimir. No additional fit parameter is introduced: the formula uses $d = 4$, α_{EM}, and $\alpha_s(\mu)$ as inputs, where α_{EM} and α_s are themselves RA-derived (Papers I–II). The identification of this formula with quark Koide breaking is a phenomenological interpolation (PI), not a theorem.

The three fermion sectors give:

Sector	Q_{N_1}	$C_{2,\text{colour}}$	K_{pred}	K_{obs}
Leptons (e, μ, τ)	3	0	0.685	0.6667 (2.8%)
Down quarks (d, s, b)	1	4/3	0.787	0.731 (7.6%)
Up quarks (u, c, t)	2	4/3	0.793	0.849 (6.6%)
Neutrinos $(\nu_e, \nu_\mu, \nu_\tau)$	0	0	2/3	2/3 (Majorana)

The $\sim$ 3–8% residuals have three sources: quark masses themselves carry $\sim 5\%$ uncertainty at the relevant scale, the RG scale choice contributes $\sim 2\%$, and equation 10 omits two-loop corrections.

The zero-free-parameter formula correctly reproduces the pattern: $K_{\rm lep} = 2/3$ exactly (no colour), $K_{\rm dn} < K_{\rm up}$ (down quarks carry smaller Q_{N_1}), and neutrinos satisfy $K = 2/3$ in the Majorana convention (§3.6).

The factor $2^{d-1}\alpha/\pi = 8\alpha/\pi$ uses identically the factor $8 = 2^{d-1}$ appearing in the Wyler formula for $\alpha_{\rm EM}$. Both expressions involve the same BDG dimension factor for $d = 4$.

3.5. The Cabibbo angle and the CKM hierarchy

The Koide parametrization fit to the charged lepton masses converges to:

$$\theta_{\rm lep} = \frac{2}{9} = 0.222222\ldots \text{ rad}, \tag{11}$$

to one part in 10^5 (the fit gives 0.222220). This is not a numerical accident: $2/9 = (2/3) \times (1/3)$, where $2/3$ is the Koide ratio K and $1/3 = 1/N_{\rm col}$. In terms of RA group-structure integers:

$$\theta_{\rm lep} = \frac{2}{N_{\rm gen} \cdot N_{\rm col}} = \frac{2}{3 \times 3} = \frac{2}{9}, \tag{12}$$

where $N_{\rm gen} = 3$ (generations from $\mathrm{SU}(3)_{\rm gen}$) and $N_{\rm col} = 3$ (colours from $\mathrm{SU}(3)_{\rm colour}$).

The three Koide sector angles form an arithmetic progression:

$$\theta_{\rm dn} = \frac{2}{9} - \delta, \quad \theta_{\rm lep} = \frac{2}{9}, \quad \theta_{\rm up} = \frac{2}{9} + \delta, \tag{13}$$

with $\delta = (\theta_{\rm up} - \theta_{\rm dn})/2 \approx 0.0665$ rad. The lepton sector sits at the central value; quarks are displaced by $\pm\delta$ due to their BDG charge coupling.

Conjecture 1 (Cabibbo angle from $\mathrm{SU}(3)_{\rm gen}$)**.** *The lepton Koide angle $\theta_{\rm lep} = 2/9$ predicts the Cabibbo angle:*

$$|V_{us}| = \sin\left(\frac{2}{N_{\rm gen} \cdot N_{\rm col}}\right) = \sin\left(\frac{2}{9}\right) = 0.2204. \tag{14}$$

The observed value is $|V_{us}| = 0.22453 \pm 0.00044$ (PDG 2022), a match to 1.8%. This identification is phenomenological (PI): the numerical proximity motivates the conjecture but does not establish it.

The Koide angle displacement. The displacement δ satisfies a Casimir-structured formula:

$$\delta = C_2(\mathrm{SU}(2)_{\rm fund}) \cdot \frac{\alpha_s(\mu)}{\pi} = \frac{3}{4} \cdot \frac{\alpha_s(\mu)}{\pi}, \tag{15}$$

where $C_2(\mathrm{SU}(2)_{\rm fund}) = 3/4$ is the quadratic Casimir of SU(2) in the fundamental representation and $\mu \approx 1.66$ GeV. Numerically: $(3/4)(\alpha_s/\pi) = 0.06660$ against the fitted $\delta = 0.06653$, a 0.1% match.

This completes a unified Casimir structure for all one-loop $\mathrm{SU}(3)_{\rm gen}$ corrections:

$$\frac{C_2(\mathrm{SU}(3))}{C_2(\mathrm{SU}(2))} = \frac{4/3}{3/4} = \frac{16}{9} = \frac{|c_{N_3}|}{|c_{N_2}|}, \tag{16}$$

connecting the group Casimirs directly to the BDG coefficient ratio.

Conjecture 2 ($|V_{cb}|$ from $\mathrm{SU}(3)_{\mathrm{gen}}$ Haar measure)**.** *The CKM element $|V_{cb}|$ satisfies:*

$$|V_{cb}| = \frac{2}{\pi}\delta = \frac{3\alpha_s(\mu)}{2\pi^2}. \tag{17}$$

Numerically: $3\alpha_s/(2\pi^2) = 0.04240$ against the observed $|V_{cb}| = 0.04221 \pm 0.00078$ (PDG 2022), a 0.4% match. As with the Cabibbo angle, this is a phenomenological identification (PI) supported by numerical proximity.

Together, the two conjectures give all leading CKM parameters from two RA quantities—$\theta_0 = 2/9$ and $\alpha_s(\mu)/\pi$:

CKM element	Formula	Error vs PDG
$\lvert V_{us}\rvert$	$\sin(2/9) = 0.2204$	1.8%
δ (angle displacement)	$(3/4)(\alpha_s/\pi) = 0.0666$	0.1%
$\lvert V_{cb}\rvert$	$(2/\pi)\delta = 0.0424$	0.4%
A (Wolfenstein)	$\lvert V_{cb}\rvert/\lambda^2 = 0.872$	4.3%

No free parameters. The $|V_{ub}|$ element and the CP-violating phase δ_{CP} require the full complex $\mathrm{SU}(2)_{\mathrm{weak}}$ vertex structure in $\mathrm{SU}(3)_{\mathrm{gen}}$ space, which remains an open target.

3.6. Neutrino masses and the Majorana identification

The $\mathrm{SU}(3)_{\mathrm{gen}}$ structure extends to the neutrino sector. The neutrino mass matrix has the same form as equation 7, with parameters m_0^ν and θ_ν specific to the neutrino sector.

The observed mass-squared differences from oscillation experiments ($\Delta m_{21}^2 \approx 7.41 \times 10^{-5}\,\mathrm{eV}^2$, $\Delta m_{31}^2 \approx 2.511 \times 10^{-3}\,\mathrm{eV}^2$, normal hierarchy, PDG 2022) fit the Koide parametrization with $m_0^\nu \approx 9.85\,\mathrm{meV}$ and $\theta_\nu \approx 0.478\,\mathrm{rad}$, giving individual masses $m_1 \approx 0.36\,\mathrm{meV}$, $m_2 \approx 8.6\,\mathrm{meV}$, $m_3 \approx 50.1\,\mathrm{meV}$, and $\sum m_\nu \approx 59\,\mathrm{meV}$.

A naive check of the Koide formula with positive square roots gives $K_\nu \approx 0.52$, apparently differing from 2/3. However, this is a category error for Majorana neutrinos. In the Koide parametrization $\sqrt{m_k} = \sqrt{m_0}(1 + \sqrt{2}\cos(\theta + 2\pi k/3))$, the relation $K = 2/3$ holds analytically because $\sum_k \cos(\theta + 2\pi k/3) = 0$ forces $\sum \mathrm{val}_k = 3$ and $\sum \mathrm{val}_k^2 = 6$, giving $K = 6/9 = 2/3$ regardless of m_0 or θ.

For the neutrino fit, one of the three values $(1 + \sqrt{2}\cos(\theta_\nu + 2\pi k/3))$ is negative (≈ -0.191 for $k = 1$). The mass $m_k = m_0\,\mathrm{val}_k^2$ is still positive (a square), but the signed square root $\sqrt{m_0}\,\mathrm{val}_k$ is negative. This sign flip is precisely a Majorana phase: for Dirac fermions all mass eigenvalues are positive-definite; for Majorana fermions the mass matrix is symmetric rather than Hermitian, and one eigenvalue carries a relative phase, appearing as a sign flip in $\sqrt{m_k}$.

When the Koide formula is evaluated using signed square roots—the correct convention for a Majorana mass matrix—both $\sum \mathrm{val}_k^2 = 6$ (unchanged by sign flips) and $\sum \mathrm{val}_k = 3$ (unchanged), giving:

$$K_{\nu,\mathrm{Majorana}} = \frac{\sum_k m_k}{\left(\sum_k \pm\sqrt{m_k}\right)^2} = \frac{6m_0}{9m_0} = \frac{2}{3}. \tag{18}$$

The apparent gap vanishes entirely. $\mathrm{SU}(3)_{\mathrm{gen}}$ holds exactly for neutrinos in the correct Majorana convention.

Proposition 4 (Structural compatibility with Majorana neutrinos)**.** *The* $\mathrm{SU}(3)_{\mathrm{gen}}$ *Koide parametrization for neutrino masses, when fit to observed mass-squared differences, requires one signed square root to be negative. A negative signed square root in the Koide parametrization is the algebraic signature of a Majorana mass eigenvalue (a relative sign in the symmetric mass matrix). Under the identification of the Koide parametrization with a physical mass matrix (PI), RA is therefore structurally compatible with—and most naturally extended to—neutrinos being Majorana particles. This is a PI-level indication supported by the algebraic fit, not an independent first-principles derivation. It is testable via neutrinoless double beta decay experiments (KamLAND-Zen, GERDA, CUORE).*

Testable mass-scale identification. The Koide parametrization with $K_\nu = 2/3$ in the Majorana convention, together with the two observed Δm^2 values, determines the absolute mass scale at PI level:

$$m_1 \approx 0.36\,\mathrm{meV}, \quad \sum_k m_{\nu_k} \approx 59\,\mathrm{meV}. \tag{19}$$

This identification is (i) well within the current cosmological bound $\sum m_\nu < 120\,\mathrm{meV}$ (Planck 2018); (ii) testable with future precision surveys (CMB-S4, Euclid, DESI) reaching sensitivity $\sum m_\nu \sim 20\,\mathrm{meV}$; and (iii) consistent with normal hierarchy ($m_1 < m_2 \ll m_3$). A positive neutrinoless double-beta-decay signal would support both the Majorana identification and the $\mathrm{SU}(3)_{\mathrm{gen}}$ framework simultaneously.

3.7. The QCD scale

The QCD confinement scale in RA is:

$$\mu_{\mathrm{QCD}} = \exp\left(\sqrt{4\Delta S^*}\right) = \exp\left(\sqrt{4 \times 0.600685}\right) \approx 4.7118, \tag{20}$$

where $\Delta S^* = -\log P_{\mathrm{acc}}(1) \approx 0.600685$ nats is the actualization threshold from Paper I (canonical value from the deterministic D4U02 enumeration, `d4u02_enumeration.py`). The fitted QCD scale parameter corresponds to $\mu_{\mathrm{QCD}} \approx 4.713$, a match to 0.03%—again with zero free parameters.

4. The Mass Cascade

Epistemic flag. This section is the most conjectural in the paper. The μ^5 gluon-vertex cascade mechanism is motivated by the BDG topology (the gluon vertex carries profile $(1, 2, 0, 0)$, which gives rise to μ^5 in the Poisson-CSG approximation), but the identification of this mechanism with the physical proton mass requires a self-consistency argument that introduces one conjectural identification: $N_{\mathrm{eff}} = L_q^3 = 64$. The numerical agreement is strong (0.3% for m_p, better for r_p and m_H), but the reader should interpret these as phenomenological interpolations (PI) in the sense of §1: motivated identifications whose numerical agreement is the principal evidence. Sharpening any of them to a first-principles derivation is an open target.

The proton mass is not input into RA. It is identified via a multiplicative cascade across the BDG depth layers, given a small number of identification steps whose epistemic status is noted as we proceed. This section gives the full construction, the self-consistency argument that fixes the operating density, and the resulting identifications for neutron, pion, Higgs, and W masses.

4.1. The μ^5 gluon vertex cascade

The proton is a three-quark bound state sustained by gluon exchange. In the BDG classification (Theorem 1), the gluon vertex carries the topological pattern $N = (1, 2, 0, 0)$ with score $S = +18$. At a local internal actualization density μ_p, the Poisson-CSG probability of forming this vertex is:

$$\mathbb{P}(\text{gluon} \mid \mu_p) = \frac{\mu_p^5}{c_4} \exp\left(-\sum_{k=1}^{4} \frac{\mu_p^k}{k!}\right) \approx \frac{\mu_p^5}{c_4} \quad (\mu_p \ll 1), \tag{21}$$

with $c_4 = 8$ the BDG fourth-layer coefficient. The exponent 5 is structural: one factor of μ from $N_1 = 1$ (one direct causal ancestor) and four factors from $\text{Pois}(\mu^2/2)$ evaluated at $k = 2$ giving $\mu^4/8$. The combination $\mu^{1+4}/(1 \cdot 2) = \mu^5/2$ times the c_4 normalization from the depth-4 layer gives $\mu^5/8$.

The proton mass equals the Planck mass suppressed by this gluon vertex rate:

$$\frac{m_p}{m_P} = \frac{\mu_p^5}{c_4}. \tag{22}$$

This is the RA mechanism for dimensional transmutation: five powers of a small density parameter produce the twenty orders of magnitude separating the proton mass from the Planck mass. No renormalization group running is invoked, and no continuum QCD input is required.

4.2. Self-consistent internal density

The proton is a self-sustaining pattern: its own actualization events create the internal density μ_p that sustains its topology. This gives a self-consistency equation:

$$\mu_p = \rho_p \cdot V_{\text{coh}} \cdot p_{\text{th}}, \tag{23}$$

where $\rho_p = N_{\text{eff}} m_p / ((4\pi/3) r_p^3)$ is the actualization density (events per 3-volume), $V_{\text{coh}} = (\pi/4!)\ell_C^4$ is the Alexandrov coherence 4-volume at the Compton wavelength $\ell_C = \hbar/(m_p c)$, $p_{\text{th}} = \alpha_{\text{EM}}$ is the per-link actualization probability identified via the optical theorem, and N_{eff} is the effective vertex count per proton cycle.

The π from the 3D spherical volume cancels exactly against the π from the 4D Alexandrov volume:

$$\frac{V_{\text{coh}}}{(4\pi/3)} = \frac{\pi/24}{4\pi/3} = \frac{1}{32}, \tag{24}$$

leaving a pure-integer geometric factor. The self-consistency equation simplifies to:

$$\mu_p = \frac{N_{\text{eff}}\, \alpha_{\text{EM}}}{32\, L_q^3}, \tag{25}$$

where we have used $r_p = L_q\, \ell_C$ with $L_q = 4$ the quark confinement depth (Lean-verified, §2).

4.3. The $N_{\text{eff}} = L_q^3$ identification

The identification $N_{\text{eff}} = L_q^3 = 64$ is introduced as a working hypothesis. It produces a universal relation $\mu_p = \alpha_{\text{EM}}/32$ that is independent of the confinement length, and gives the observed proton

mass to 0.3%. The identification is conjectural (CN in the sense of §1): its motivation is structural—the proton's confinement region should contain L_q^3 independent actualization sites in depth-space, corresponding to the 3D volume of the confinement lattice—but this is a physical picture rather than a derivation from the growth rule.

With this identification, the confinement length drops out of the self-consistency equation:

$$\mu_p = \frac{\alpha_{\mathrm{EM}}}{32}. \tag{26}$$

Substituting into equation (22):

$$m_p = m_P \cdot \frac{\alpha_{\mathrm{EM}}^5}{2^{28}} = 941\,\mathrm{MeV}. \tag{27}$$

The observed proton mass is 938.3 MeV; the match is 0.3%. The exponent $28 = 5\times5+3$ decomposes as five powers of $32 = 2^5$ from equation (26) plus $c_4 = 2^3$ from equation (22). The inputs are m_P (the single dimensionful scale in RA), $\alpha_{\mathrm{EM}} = 1/137.036$ (derived in §3.2), and the conjectural identification $N_{\mathrm{eff}} = L_q^3$.

Remark 3 (On $N_{\mathrm{eff}} = 64$)**.** *The identification $N_{\mathrm{eff}} = L_q^3$ is motivated by two observations: (i) this identification makes μ_p independent of L_q, giving the universal relation $\mu_p = \alpha_{\mathrm{EM}}/32$ for all hadrons bound by the same gluon vertex; (ii) the resulting 0.3% match to the proton mass is tighter than would be expected by accident given the cascade's five-power amplification of N_{eff}. Neither of these is a derivation. Deriving N_{eff} from first principles—for instance, from the BDG Poisson-CSG self-consistency of a confined quark-gluon region—remains the primary open target of the cascade programme.*

4.4. The $d = 4$ uniqueness of the cascade exponent

The cascade exponent $1 + 2N_2 = 2N_c - 1$ arises from the gluon vertex requiring $N_1 = 1$ and $N_2 = N_c - 1 = 2$, where $N_c = 3$ is the number of colors. The confinement cycle length is $d+1$. The identity:

$$2N_c - 1 = d + 1 \iff d = 4 \tag{28}$$

holds only in four spacetime dimensions. This is a new uniqueness argument for $d = 4$, independent of the D4U02 selectivity-ceiling argument given in Paper I. In other dimensions, the cascade either fails to produce dimensional transmutation ($d < 4$: exponent too small) or overshoots ($d > 4$: exponent too large, producing masses above the observed hadronic scale).

4.5. Neutron, pion, and related identifications

The same cascade mechanism, applied to mesons and to the neutron, yields three additional identifications with no new parameters.

Neutron mass. The neutron has identical BDG topology to the proton (both are three-quark systems bound by gluon exchange at the same density μ_p). The cascade formula predicts $m_n = m_p$ to leading order. The observed splitting $\Delta m = m_n - m_p = 1.293\,\mathrm{MeV}$ is an electromagnetic correction from the u-d charge difference, consistent with standard QCD and within the 0.3% cascade accuracy.

Pion mass. A meson is a two-quark system. If N_{eff} scales with quark count, then mesons have $N_{\text{eff}}^{\text{meson}} = (2/3) \times 64 = 42.7$, giving:

$$\mu_\pi = \frac{2}{3}\mu_p, \quad m_\pi = \left(\frac{2}{3}\right)^5 m_p = \frac{32}{243} \times 941\,\text{MeV} \approx 124\,\text{MeV}. \tag{29}$$

The observed charged pion mass is $m_{\pi^\pm} = 139.6\,\text{MeV}$ (11% discrepancy). The residual is discussed in §4.6. Non-strange vector mesons (ρ, ω) have $N_{\text{eff}} \approx 62$ and identified masses close to m_p, consistent with observation.

Flavor-singlet pseudoscalar (η'). Among the light pseudoscalar mesons, the η' is distinguished by its flavor-singlet structure: in the Standard Model, it is the pseudoscalar whose mass is not suppressed by the Goldstone mechanism due to the $\text{U}(1)_A$ anomaly. Its observed mass is $m_{\eta'} = 957.8\,\text{MeV}$. The baryon cascade formula predicts 941 MeV, matching $m_{\eta'}$ to 1.8% with no free parameters and without invoking the quark-count scaling of equation (29). We flag this as a separate phenomenological identification (PI): the η' appears to track the baryon scale because, in RA as in the Standard Model, it lacks the Goldstone-like suppression that distinguishes π, K, η. This match was found in post-hoc analysis of the cascade and is reported here for completeness; it was not used to tune any RA parameter.

Proton charge radius. The confinement radius follows directly from the Compton-wavelength identification:

$$r_p = L_q \cdot \ell_C = L_q \cdot \frac{\hbar}{m_p c}. \tag{30}$$

Using $L_q = 4$ and the RA-identified $m_p = 941\,\text{MeV}$: $r_p = 0.839\,\text{fm}$. Using the observed $m_p = 938.3\,\text{MeV}$: $r_p = 0.841\,\text{fm}$. The CODATA value is $r_p = 0.8414 \pm 0.0019\,\text{fm}$—a match to 0.03% with the observed mass and 0.3% with the cascade identification. As with the other mass-cascade results, this should be read as a phenomenological interpolation: the numerical proximity is encouraging, but the identification $r_p = L_q \cdot \ell_C$ is not yet derived from first principles.

4.6. The pion residual and chirality-from-directionality

The 11% residual in the pion cascade identification is the most visible gap in §4. We do not close it in this paper, but we identify where an RA-native account of the gap should be sought.

What we do not have. We do not have an RA derivation of a Goldstone mechanism. In the Standard Model, the pion is a pseudo-Goldstone boson of spontaneously broken chiral $\text{SU}(2)_L \times \text{SU}(2)_R$ symmetry, with mass generated by explicit breaking from nonzero light-quark masses (the Gell-Mann–Oakes–Renner relation). This derivation uses Dirac spinors, a continuous chiral symmetry, and a Lagrangian; none of these are RA primitives. Importing the mechanism directly into RA would not be a derivation but an identification.

What we do have. In RA, chirality is not a symmetry that gets broken. The underlying graph is a DAG (Axiom 1, Paper I), which means time (graph progression) is structurally unidirectional rather than spontaneously broken. There was never a symmetric starting point. This suggests the

RA-native analog of chiral symmetry breaking is not a broken symmetry at all, but a structural asymmetry built into the ontology of actualization events. The BDG coefficients $(1, -1, 9, -16, 8)$ display a striking sign-alternation by chain-length parity; whether this parity structure carries the chiral content of RA dynamics is an open question, requiring careful distinction from the Möbius-function structure on causal intervals that the coefficients formally implement.

What might close it. Three kinds of work would be needed to convert the 11% gap into a derivation rather than a phenomenological acknowledgment:

1. A precise RA-native definition of a "meson" configuration in the growing DAG—i.e., a specification of which 2-vertex bound state corresponds to the pion—distinct from the 3-vertex baryon configuration used in §4.
2. A mechanism by which DAG directionality generates a mass scale for this 2-vertex configuration that is lighter than the baryon scale in a structurally derivable way (rather than by $(2/3)^5$ quark-count scaling).
3. Either (a) an RA-native Goldstone theorem showing that a specific symmetric excitation of the 2-vertex configuration has suppressed mass, with a combinatorial suppression factor computable from BDG coefficients; or (b) an alternative mechanism that abandons the Goldstone framing entirely.

Option (3b) is worth emphasizing. The Goldstone framing is inherited from continuum quantum field theory, and it is not obvious that RA should reproduce it. The 11% residual could instead be the signature of a different RA-native mechanism that has no Standard Model analog—one which gives the same leading scale as Goldstone suppression but with a different structural origin. Ruling between these options is a target for future work.

Status. Paper II's pion identification remains at 124 MeV (11% low) via equation (29). We do not upgrade this in the present suite. Post-hoc Monte Carlo exploration of the BDG dynamics at $\mu = 1$ revealed numerical regularities (for instance, $m_{\pi^0}^2 \approx m_p^2/48$ at 0.6% error) that are suggestive of finer structure, but no derivation has been produced. These regularities are recorded in the RA knowledge base as conjectures (CN) pending derivation work, and are not claimed here.

4.7. Higgs mass

The Higgs boson carries the scalar depth-2 pattern (N_2-dominant). Its mass is related to the proton mass by the electromagnetic depth-ratio factor:

$$\frac{m_H}{m_p} = \alpha_{\mathrm{EM}}^{-1} - L_q = 137 - 4 = 133. \tag{31}$$

Substituting the RA-identified m_p: $m_H = 133 \times 941\,\mathrm{MeV} = 125.2\,\mathrm{GeV}$. The observed value is $m_H = 125.25 \pm 0.17\,\mathrm{GeV}$—a match to 0.06%, well within experimental uncertainty.

The ratio $m_H/m_p = 133$ is a phenomenological identification (PI) within the BDG framework: no additional free parameter is introduced, but the identification of the Higgs with a depth-2 scalar pattern whose mass scale is set by the electromagnetic depth-ratio (137, from Paper I) reduced by the quark confinement depth (4, from the topology closure) is not yet a first-principles derivation. The numerical proximity (0.06%) is encouraging but does not by itself elevate the identification to theorem-level status.

4.8. W boson mass

The W boson carries the depth-1 gauge pattern with net weak isospin (Type 3 in Theorem 1). Its mass is constrained by the BDG debt structure of the weak charge, but the exact derivation involves the N_3/N_4 handedness sector and is not yet closed. The provisional formula:

$$m_W = 6^{5/2}\, m_p \approx 83\,\text{GeV} \tag{32}$$

matches the observed 80.4 GeV to 3.3%, but the exponent 5/2 has not been derived from BDG structure. This is labeled as a conjecture; the Z boson and W boson masses together are a target for the handedness-sector derivation.

4.9. Baryon-to-dark matter ratio

The ratio of non-baryonic to baryonic gravitational weight is derived from BDG path weight enumeration. The baryon weight is exact:

$$W_{\text{baryon}} = W_{\text{quark}} + W_{\text{fermion}} = \frac{1}{2!1!} + \frac{1}{1!1!1!1!} = \frac{1}{2} + 1 = \frac{3}{2}. \tag{33}$$

The non-baryon weight is computed by BDG exact enumeration over all non-baryon **N**-vectors with $S_{\text{BDG}} > 0$:

$$W_{\text{other}} = 25.97 \quad \text{(exact enumeration, max-}n = 12,\ \text{stable)}. \tag{34}$$

The ratio:

$$\frac{W_{\text{other}}}{W_{\text{baryon}}} = \frac{25.97}{1.5} = 17.32. \tag{35}$$

The energy weighting at the hadronic-partonic crossover scale $Q_{\text{eff}} = 2m_p$ (the $B\bar{B}$ pair threshold, the natural IR scale of QCD nuclear physics) uses $\alpha_s(2m_p) = 0.312$ from standard SM QCD RG running:

$$f_0 = 17.32 \times \alpha_s(2m_p) = 17.32 \times 0.312 = 5.40. \tag{36}$$

The Planck satellite measurement gives $\Omega_{\text{DM}}/\Omega_b = 5.416 \pm 0.015$—a match to 0.3%.

Epistemic note. The 17.32 ratio is fully RA-native: it is derived from BDG exact enumeration with zero free parameters. The $\alpha_s(2m_p) = 0.312$ factor is *not* RA-native: it uses established SM QCD RG running from the UV fixed point $\alpha_s(m_Z) = 1/\sqrt{72}$ (which IS RA-native, §3.1) down to the hadronic scale. This RG-running step is not a free parameter, but it *is* an external input to the f_0 calculation. A future RA-native derivation of the $\alpha_s(\mu)$ profile between the BDG UV and IR fixed points would close this remaining bridge.

5. Force Ranges: One Principle, Three Outcomes

The range of a force in RA follows from a single unified principle: how far an exchange pattern can propagate before its actualization cost per unit length exceeds what the local causal bandwidth can sustain. This gives three qualitatively distinct outcomes—Coulombic, Yukawa-like, and confining—from the same BDG structure.

Force	Exchange	BDG debt	Range mechanism
EM	photon	0	$1/r^2$ from causal 2-sphere geometry
Weak	W, Z	> 0	Lifetime $\hbar/m_W c^2$; range $\sim 10^{-18}$ m
Strong	gluon	0 (colored)	LLC flux tube; linear $V(r) = \sigma r$

5.1. Electromagnetism: $1/r^2$ from causal geometry

The photon carries zero BDG debt per propagation cycle ($\Delta S_{\text{BDG}} = 0$) and zero net spatial charge ($Q_{N_2} = 0$): it propagates with zero actualization cost and no LLC imbalance. Its exchange pattern spreads freely in all directions.

The $1/r^2$ fall-off is the geometry of the causal light cone: the number of causal edges crossing a 2-sphere of coordinate radius r scales as r^2, so the photon exchange density—and therefore the force—falls as $1/r^2$. Coulomb's law is a consequence of the 4D causal structure, not a separate postulate.

5.2. The weak force: range from BDG debt

The W and Z bosons carry net $Q_{N_3/N_4} \neq 0$ (weak isospin), paying BDG actualization cost at every vertex of their propagation. A pattern with rest-frame actualization frequency $f_0 = m_W c^2/h$ can persist for proper time $\tau_W = 1/f_0 = \hbar/(m_W c^2)$ before its BDG debt becomes unsustainable. The range is:

$$R_W = c\,\tau_W = \frac{\hbar}{m_W c} \approx 2 \times 10^{-18}\,\text{m}. \tag{37}$$

No Yukawa mechanism is postulated; this is the actualization lifetime of any BDG-indebted pattern. The weak force is short-range because the W and Z carry handedness charge that must be continuously actualized.

5.3. The strong force: asymptotic freedom and confinement

The strong force has two apparently contradictory properties: at short distances quarks behave as nearly free particles (asymptotic freedom), at long distances they are permanently confined. In RA both emerge from the same Poisson-CSG density parameter $\mu(x)$ varying with the actualization density between the quarks.

Asymptotic freedom at high μ. At short quark separation, the causal region between them is densely actualized. The N_2 spatial branching charge of each quark is distributed across many intermediate causal vertices: the effective colour charge experienced by a nearby quark is diluted over $\sim e^{\mu}$ available causal paths. In Poisson-CSG language, the mean N_2 branching per vertex spreads uniformly over all available paths, so the effective colour charge per path falls as $e^{-\mu}$. At high μ (short distance, high energy), colour charges decouple: $\alpha_s \to 0$. This is asymptotic freedom, derived from the Poisson-CSG density dependence rather than inserted by hand.

Confinement and the LLC flux tube at low μ. As the quarks are separated, the causal structure between them becomes sparser. The LLC requires $Q_r + Q_g + Q_b = 0$ globally at all times, but with fewer intermediate vertices, global LLC balance must be maintained by fewer causal paths.

Those paths must each carry higher N_2 branching flux to compensate. The exchange pattern does not spread over a 2-sphere as it would for a massless, colourless particle; instead it collimates into a narrow tube of concentrated N_2-density causal structure: a QCD flux tube.

Each additional unit of quark separation requires one additional layer of N_2 causal structure in the flux tube to maintain LLC balance. The actualization cost per unit length is therefore constant:

$$V(r) = \sigma \cdot r, \quad \sigma = \text{string tension (constant)}. \tag{38}$$

The force $F = dV/dr = \sigma$ is constant and independent of separation: the strong force does not weaken with distance but maintains a constant tension. This is the RA-native explanation of quark confinement without any phenomenological input.

Flux tube snap and hadron production. When the quarks are sufficiently separated, the accumulated BDG action of the stretched flux tube exceeds the actualization cost of nucleating a new quark-antiquark pair at the snap point:

$$\sigma \cdot r > 2m_q c^2 \implies \text{flux tube snaps, new meson nucleates at each end.} \tag{39}$$

This is why isolated quarks are never observed: every attempt to separate them nucleates new hadrons at the snap point.

5.4. Regge trajectories as a bonus result

In QCD, hadrons of the same quantum numbers but different spins lie on linear Regge trajectories: $m^2 \propto J$. This is a striking experimental regularity that QCD explains via the flux tube but does not derive from first principles.

In RA, a baryon with orbital angular momentum J has a flux tube of length $r \propto \sqrt{J}$ (from $J \propto mvr$ at the string tension scale). The mass is:

$$m \propto \sigma \cdot r \propto \sigma^{1/2}\sqrt{J}, \quad m^2 \propto \sigma J. \tag{40}$$

Regge trajectories are a direct consequence of the LLC flux tube picture; they require no additional input beyond the linear potential of equation (38).

5.5. The BDG renormalization group: two exact fixed points

The qualitative picture above can be made quantitative via a BDG-native renormalization group. At density μ, the effective BDG score of a Type-3 (quark) vertex interpolates between the confinement limit (dominated by the vacuum score c_0) and the asymptotic-freedom limit (dominated by the free-vertex score c_4):

$$S_{\text{eff}}(\mu) = (1 - e^{-\mu})c_4 + e^{-\mu}c_0. \tag{41}$$

The strong coupling at density μ is:

$$\alpha_s(\mu) = \frac{1}{\sqrt{c_2\, S_{\text{eff}}(\mu)}}. \tag{42}$$

This model has two exact fixed points determined entirely by the BDG integers $(c_0, c_2, c_4) = (1, 9, 8)$:

$$\alpha_s^{\mathrm{IR}} = \lim_{\mu\to 0} \alpha_s(\mu) = \frac{1}{\sqrt{c_2\, c_0}} = \frac{1}{3} \quad \text{(confinement)}, \tag{43}$$

$$\alpha_s^{\mathrm{UV}} = \lim_{\mu\to \infty} \alpha_s(\mu) = \frac{1}{\sqrt{c_2\, c_4}} = \frac{1}{\sqrt{72}} = 0.11785 \quad \text{(asymptotic freedom)}. \tag{44}$$

The IR fixed point corresponds to the dense, strongly-coupled confinement regime; the UV fixed point to the sparse, weakly-coupled asymptotic-freedom regime. Both are exact consequences of the BDG integers with no external parameters. The physically observed running of α_s from the hadronic scale ($\alpha_s \sim 0.3$) to the electroweak scale ($\alpha_s(m_Z) \approx 0.118$) is bracketed by these two fixed points.

Observable	RA prediction	Experiment	Match	Status
$\alpha_s(m_Z)$	$1/\sqrt{72} = 0.11785$	0.1180 ± 0.0009	0.13%	CV
$\alpha_{\mathrm{EM}}^{-1}$	137.036019	137.035999	0.00001%	CV
m_p	941 MeV	938.3 MeV	0.3%	PI
m_H	125.2 GeV	125.25 GeV	0.06%	PI
r_p	0.84 fm	0.841 fm	0.03%	PI
f_0	5.42	5.416	0.07%	CV (w/ ext. RG)
μ_{QCD}	4.7118	4.713 (fit)	0.03%	CV
K (Koide)	2/3	$2/3 \pm 10^{-5}$	exact	LV

6. The Gauge Group from BDG Geometry

The Standard Model gauge group SU(3) × SU(2) × U(1) is identified in this paper, at the DR/PI level, with the product of two independent geometric structures on the BDG-filtered growth process. The identification is a programme-level proposal, not a closed theorem chain from BDG primitives to a fully derived gauge group.

6.1. Inter-depth geometry and SU(3)

The BDG coefficients alternate in sign: $(+1, -1, +9, -16, +8)$. This sign alternation arises from inclusion-exclusion counting of causal intervals in $d = 4$ [1]. Positive coefficients distribute isotropically (full symmetry at that depth); negative coefficients concentrate anisotropically (broken symmetry).

The inter-depth transfer matrices $T_{i\to j}$ describe how probability flows between BDG depth sectors i and j during the sequential growth process. These matrices carry *phase kicks* $\Delta S_{ij} = c_j - c_i$ that are profile-independent:

Transition	c_i	c_j	Phase kick ΔS
$1 \to 2$	-1	$+9$	$+10$
$2 \to 3$	$+9$	-16	-25
$3 \to 4$	-16	$+8$	$+24$
$1 \to 3$	-1	-16	-15
$2 \to 4$	$+9$	$+8$	-1
$1 \to 4$	-1	$+8$	$+9$

The magnitudes $|10|, |25|, |24|, |15|, |1|, |9|$ are all distinct, which is the condition for geometric (Berry) phase.

6.2. The Berry phase programme

Theorem 2 (Transfer-Graph Overlap)**.** *For Givens-type sector transfers $T_{i\to j}$ and $T_{k\to l}$ acting on the BDG sector space, $[T_{i\to j}, T_{k\to l}] = 0$ if and only if $\{i, j\} \cap \{k, l\} = \emptyset$. Verified exhaustively by direct computation for $n = 2, 3, 4, 5, 6$ in* `berry_theorems.py`*. The Berry geometry is therefore controlled by the incidence structure of the sector transfer graph.*

Theorem 3 (Exact Poisson-thinning transfer law)**.** *The conditional probability that a boundary-event promotes the local motif to depth-k sector is $f_k = \lambda_k / \sum_j \lambda_j$ exactly (not approximately), where $\lambda_k = \mu^k/k!$ are the Poisson-CSG rates at density μ. This is a consequence of the Poisson thinning property: in a Poisson process partitioned into regions with rates $\{\lambda_k\}$, a uniformly random point falls in region k with probability exactly $\lambda_k / \sum_j \lambda_j$, and region counts are independent Poisson variables. The BDG transfer fractions therefore require no modeling assumption beyond the Poisson-CSG framework of Paper I.*

Theorem 4 (2-sector flatness)**.** *For any 2-sector BDG transfer system, the Berry phase is identically zero. By Theorem 2, the two transfers overlap on both indices; the resulting rotations act about the same axis in the relevant* SU(2) *Lie algebra, and same-axis rotations commute. The Wilson-loop eigenphase equals the sum of the individual rotation phases—a dynamical total phase, not a geometric one.*

Theorem 5 (Minimum curved sector count = 3)**.** *Nonzero Berry phase requires a minimum of 3 BDG depth sectors. The three BDG phase kicks around any 3-cycle of the sector graph sum to zero but are unequal in magnitude (all six pairs $|c_j - c_i|$ for the BDG coefficients $(c_1, c_2, c_3, c_4) = (-1, 9, -16, 8)$ produce six distinct values $|10|, |25|, |24|, |15|, |1|, |9|$); by Theorem 2 the adjacent transfers overlap on one index and do not commute. Combined with Theorem 4, this establishes that 3 is the minimum sector count for genuine geometric phase.*

Theorem 6 (Depth-3 curvature maximum)**.** *The strongest Berry curvature contributions come from transitions involving depth $k = 3$ (coefficient $c_3 = -16$, the largest in magnitude), producing kicks $|\Delta_{23}| = 25$ and $|\Delta_{34}| = 24$. Since $|c_3|$ exceeds the next-largest $|c_k|$ by ≥ 7, any plaquette involving depth 3 carries curvature strictly larger than any plaquette not involving depth 3.*

The 3-sector structure at the minimum required for geometric phase is identified with SU(3). The inter-depth geometry of the BDG coefficients—through their sign alternation and magnitude

asymmetry—is the candidate RA-native source of the color gauge group. The identification of the Berry-phase structure with the continuum SU(3) gauge group is a programme-level step (DR/PI), not a derivation at theorem level.

Computation-verified eigenphase. At $\mu_{\mathrm{QCD}} = 4.7118$ (canonical value from `d4u02_enumeration.py`), the 3-sector Wilson loop over the depth-$\{1, 2, 4\}$ transfer graph has eigenphases $\{-0.8055, 0.0000, +0.8055\}$ rad, computed explicitly in `berry_gauge.py` via transfer fractions from Theorem 3 and phase kicks $\Delta_{12} = +10$, $\Delta_{24} = -1$, $\Delta_{41} = -9$ from Theorem 2. The eigenphase $\varphi = 0.806$ rad is robust to the ΔS^* refinement at the 10^{-4} rad level (the transfer fractions depend on ratios of λ_k, which cancel the μ_{QCD} shift to leading order).

6.3. Intra-depth geometry and SU(2)

Orthogonal to the inter-depth structure, each BDG depth layer has *spatial* boundary directions—the directions perpendicular to the causal chain within a single depth shell. In $d = 4$, each depth shell has 2 spatial boundary directions. The symmetry group of 2-direction rotations is SU(2).

6.4. The geometric bifurcation

$$\boxed{\mathrm{SU}(3)_{\mathrm{depth}} \times \mathrm{SU}(2)_{\mathrm{direction}} \times \mathrm{U}(1)_{\mathrm{phase}} \leftrightarrow \text{SM gauge group.}} \tag{45}$$

In this proposal, the gauge group is the product of two independent geometric structures: inter-depth transfer (yielding SU(3) via the Berry-phase programme) and intra-depth boundary directions (yielding SU(2)), with the overall U(1) phase. Both structures are motivated by the BDG integers and the dimension $d = 4$. The reader is cautioned that the Berry-phase theorems above are statements about the BDG transfer-matrix structure specifically; the identification of that structure with the continuum Standard-Model gauge group is an interpretive step (DR/PI), not a derivation from BDG primitives to the continuum group at theorem level.

7. The Absence of Supersymmetry

Supersymmetry (SUSY) was motivated by three problems in the Standard Model. Within RA, each of those three problems receives a framework-internal reading that does not require supersymmetric partners to resolve it, and the structural algebra that supersymmetry requires is not naturally generated by the BDG growth dynamics. This does not constitute a proof that superpartners do not exist; it is a statement about which structural motivations for them have RA-native alternatives. The continued non-observation of superpartners at the LHC is consistent with the RA picture, though of course consistency with LHC data is also a feature of many other non-SUSY frameworks.

7.1. The hierarchy problem: dissolved

The hierarchy problem asks why quantum loop corrections do not drive the Higgs mass to the Planck scale. In QFT, virtual particle loops generate corrections $\delta m_H^2 \sim \Lambda_{\mathrm{UV}}^2$, requiring fine-tuned cancellations to maintain $m_H \sim 125$ GeV.

In RA, mass is $f_0 = mc^2/h$—the actualization frequency of a BDG pattern. There are no virtual loops in the QFT sense. Feynman diagram loops are a continuum approximation to the

causal graph, and in a DAG they cannot loop back on themselves by definition. The hierarchy problem is a consequence of assuming QFT loop corrections are real physical processes, rather than approximation artifacts of treating a discrete structure as continuous. In the discrete RA framework, the Higgs pattern simply has the BDG topology it has; there is no mechanism to drive it to a different mass.

7.2. Dark matter: already resolved

SUSY's lightest supersymmetric particle (LSP)—the neutralino—was the leading dark matter candidate for decades. In RA, the WIMP prohibition (Paper III) establishes that any particle that does not participate in on-shell actualization events contributes zero causal depth and zero metric weight. WIMPs, neutralinos, and axions are structurally excluded as gravitational sources.

The dark matter phenomena are explained by the Poisson-CSG dimensional reduction in sparse galactic halos ($\mu < 1$), developed in Paper III. The dark matter problem is not a deficit to be filled by a new particle; it is a signal of topology change in the causal graph.

7.3. Gauge coupling unification: handled at the source

SUSY modifies the running of the three gauge couplings so that they unify at a high scale $\sim 10^{16}$ GeV. In RA, the coupling constants $\alpha_{\mathrm{EM}}, \alpha_s, G_F$ are not free parameters to be unified—they are identified with properties of the BDG coefficients and the Poisson-CSG asymptotic c_k distribution. If the three couplings have specific values because they are properties of the four BDG degrees of freedom in 4D, the unification question is answered at the level of the underlying topology, not by a symmetry imposed on top of the Standard Model (see §8).

7.4. Why SUSY algebra is not generated

In the RA pattern framework, fermions are 2-periodic BDG patterns and bosons are 1-periodic. SUSY would require that for every 2-periodic pattern there exists a corresponding 1-periodic pattern with the same mass and charges.

The BDG growth dynamics do not generate this correspondence. A 2-periodic pattern is structurally different from a 1-periodic one: they have different (N_1, N_2, N_3, N_4) profiles and therefore different actualization frequencies. There is no symmetry transformation in the BDG algebra that maps a 2-periodic pattern to a 1-periodic pattern with equal eigenvalues. SUSY is not ruled out by RA—it is simply not generated. It is unnecessary structure, and the LHC's non-observation of superpartners across the entire originally motivated parameter space is consistent with this structural picture.

8. Grand Unification: $\mathbf{SU(3)}_{\mathrm{colour}} \times \mathbf{SU(3)}_{\mathrm{gen}}$

Epistemic flag. This section presents a structural conjecture (CN in the sense of §1). The identification of colour and generation as two facets of a single BDG structure at the Planck scale is motivated by the integer architecture of the framework but is not derived from first principles. The section should be read as proposing a unification picture consistent with the framework, not as establishing one. The specific value $\sin^2\theta_W = 3/8$ at unification follows if the unification holds; it is not independent evidence for it.

This section proposes that Grand Unification emerges as a natural structural consequence of the BDG architecture, rather than being an additional postulate.

8.1. Two SU(3) groups from a single source

Both $\mathrm{SU}(3)_{\mathrm{colour}}$ and $\mathrm{SU}(3)_{\mathrm{gen}}$ arise from the same BDG structure:

$$\mathrm{SU}(3)_{\mathrm{colour}}: \text{ acts on } \{r, g, b\} = \text{three spatial directions of } N_2 \text{ branching}, \tag{46}$$

$$\mathrm{SU}(3)_{\mathrm{gen}}: \text{ acts on } \{N_2, N_3, N_4\} = \text{three BDG excitation levels}. \tag{47}$$

In the macroscopic low-energy regime, these are distinguishable: spatial N_2 directions couple to colour-charged patterns, while excitation levels couple to generation structure. But at the Planck scale, where $\mu \to \infty$ and all BDG levels are equally and densely populated, the distinction between "which spatial direction" and "which excitation level" disappears. The two SU(3) groups become indistinguishable facets of a single symmetry group acting on the full six-dimensional BDG charge space $\{r, g, b, N_2, N_3, N_4\}$.

Conjecture 3 (BDG Grand Unification)**.** *At the Planck scale ($\mu \to \infty$), the combined symmetry group of the RA causal graph is* $\mathrm{SU}(3)_{\mathrm{colour}} \times \mathrm{SU}(3)_{\mathrm{gen}}$*, embedded in a unified group* G_{GUT} *acting on all six non-gravitational BDG degrees of freedom. This unification is structural—the Planck scale is not a free parameter but the scale at which the spatial and excitation-level aspects of the BDG structure become indistinguishable, i.e.,* $\mu \to \infty$ *and* $c_k \to$ *uniform.*

8.2. The Weinberg angle at unification

If $\mathrm{SU}(3)_{\mathrm{colour}} \times \mathrm{SU}(3)_{\mathrm{gen}}$ unifies into a single group at the Planck scale, the electroweak mixing angle takes the SU(5)-type value at unification:

$$\sin^2\theta_W\Big|_{\mathrm{GUT}} = \frac{3}{8}. \tag{48}$$

The observed value $\sin^2\theta_W \approx 0.231$ at the Z mass scale is the result of renormalization group running from the Planck scale to the electroweak scale. In RA, this running is the μ-dependence of the Poisson-CSG coupling: as μ decreases from ∞ (Planck) to the QCD confinement scale, the N_2 coupling strength grows relative to N_1 and N_3/N_4—consistent with the observed running of α_s that drives $\sin^2\theta_W$ from 3/8 to 0.231.

8.3. Why RA grand unification needs no SUSY

In conventional SU(5) GUTs, the three coupling constants do not quite meet without supersymmetric partners adjusting the RG flow. In RA, the unification at the Planck scale is structural: it is the $\mu \to \infty$ limit of the Poisson-CSG, not a coincidence of coupling trajectories. No superpartners are needed because the unification scale is fixed by the BDG structure, not by RG running. The non-observation of superpartners at the LHC (§7) is therefore both consistent with and explained by this picture.

8.4. Exact proton stability

In SU(5) GUTs, proton decay is predicted via X boson exchange between colour and generation sectors. In RA, the colour and generation sectors are unified at the Planck scale, but the LLC (Lean-verified) exactly conserves baryon number at every scale. The SU(3) unification does not introduce baryon-number-violating operators because the LLC constraint applies to every vertex in the DAG at every scale.

Proposition 5 (Proton stability)**.** *Baryon number is exactly conserved in RA. A baryon is a pattern carrying net N_2 spatial winding number. The LLC applied to all three spatial components of N_2 forbids any process that changes the net N_2 winding number without a corresponding anti-baryon creation or annihilation. Proton decay is structurally forbidden—not merely suppressed by a heavy X boson, but categorically prohibited by the same conservation law that confines quarks.*

This prediction is testable by next-generation neutrino/proton decay experiments (Hyper-Kamiokande, DUNE). A single observed proton decay event would falsify RA.

9. Renewal Motifs and Stability

Definition 1 (Local topology class)**.** *A local topology class is an equivalence class of finite neighborhoods in the actualized graph, classified by BDG depth profile and induced causal comparability pattern.*

Definition 2 (Renewal motif)**.** *A renewal motif is a local topology class M such that, conditioned on admissible growth in its neighborhood, the probability that the successor neighborhood remains in M is bounded away from zero.*

Stability is a property of repeated finite renewal, not eternal persistence (Axiom 7, Paper I). A particle "exists" insofar as its local topology class reproduces itself under the sequential growth rule.

Proposition 6 (Structural fragility — Lean verified)**.** *Electrons and photons have BDG stability score 1 (the minimum). They are the most fragile stable motifs—any additional interaction pushes them past the renewal threshold.*

Verified in `RA_D1_Proofs.lean` (`structural_fragility`). This is the structural basis for the Kinematic Coherence Bound in quantum computing (Paper I, §12).

9.1. The force hierarchy

The BDG depth channels provide a natural hierarchy of interaction strength. Each depth k contributes a factor $|c_k|/k!$ to the interaction amplitude:

Depth k	$\lvert c_k\rvert$	$\lvert c_k\rvert/k!$	Interaction
1	1	1.000	Strong (direct)
2	9	4.500	Electromagnetic
3	16	2.667	Weak
4	8	0.333	Residual strong / gravity

The factorial suppression $1/k!$ at each depth level produces the observed force hierarchy. The "weakness" of gravity relative to the other forces is not a fine-tuning problem but a consequence of depth-4 factorial suppression: $|c_4|/4! = 8/24 = 1/3$.

9.2. Charge quantization in units of $e/3$

Charge quantization in units of $e/3$ is a theorem about N_2 winding numbers in the BDG closure pattern. In a 3+1D causal graph, a vertex can receive signed edges from at most three independent spatial directions. The net signed N_1 charge is therefore constrained to integer values in $\{-3, -2, -1, 0, +1, +2, +3\}$. Identifying the unit charge with $e/3$ (one spatial edge per dimension):

$$Q_{N_1} \in \{-e, -2e/3, -e/3, 0, +e/3, +2e/3, +e\}. \tag{49}$$

This matches the complete set of electric charges in the Standard Model: up quark $+2e/3$ ($Q_{N_1} = +2$), down quark $-e/3$ ($Q_{N_1} = -1$), electron $-e$ ($Q_{N_1} = -3$), neutrino 0 ($Q_{N_1} = 0$), $W^{\pm}$ ($Q_{N_1} = \pm 3$), and neutral particles ($Q_{N_1} = 0$). Electric charge is quantized in units of $e/3$ because the spatial dimensionality of the causal graph is exactly 3.

10. Further Consequences

The machinery developed in the preceding sections offers RA-native readings of three additional long-standing problems in physics. In each case the reading is not a full solution in the traditional sense but a reframing: the framework's discrete primitives remove the need for certain questions to be asked, because what looked like a puzzle in the continuum picture turns out to be an artifact of the continuum description. We present three here.

10.1. The strong CP problem: $\theta_{\mathrm{QCD}} = 0$ exactly

The QCD Lagrangian permits a CP-violating term $\mathcal{L}_\theta = \theta_{\mathrm{QCD}}(g^2/32\pi^2)G^{\mu\nu}\tilde{G}_{\mu\nu}$. The experimental bound $|\theta_{\mathrm{QCD}}| < 10^{-10}$ implies this term is extraordinarily small, but no symmetry of the Standard Model forces it to zero. This is the strong CP problem.

The θ_{QCD} term arises in QFT because the QCD vacuum is a superposition of topological sectors $|n\rangle$ indexed by the winding number $n \in \pi_3(\mathrm{SU}(3)) = \mathbb{Z}$. These sectors are connected by instantons: Euclidean field configurations on $\mathbb{R}^4$ with non-trivial topology, which contribute to the path integral as tunnelling amplitudes between sectors.

In RA, the vacuum is the unique empty graph with no vertices. The growing DAG has no smooth gauge-field topology and therefore no concept of winding-number sectors: π_3 of a discrete partially-ordered set is trivial. There is no superposition of $|n\rangle$ states, because the sequential, acyclic growth of the graph admits only one vacuum—the empty graph. Instantons, which require a smooth Euclidean field configuration on $\mathbb{R}^4$ with non-trivial homotopy, have no counterpart in a discrete DAG.

Proposition 7 (Strong CP dissolution)**.** *The QCD vacuum's topological structure—a superposition of winding-number sectors connected by instantons—requires a smooth Euclidean gauge field with non-trivial* $\pi_3(\mathrm{SU}(3))$ *topology. The RA causal graph has a unique empty vacuum with no such*

topological structure. The mechanism that generates θ_{QCD} is absent, so $\theta_{\mathrm{QCD}} = 0$ exactly. The strong CP problem does not arise in RA.

A second, independent argument reaches the same conclusion via the LLC. The strong force is the LLC constraint on N_2 spatial charge. CP transformation flips Q_{N_2} and reverses the spatial branching direction. The LLC equation $\Delta Q_{N_2} = 0$ at every vertex is equivalent to $\Delta(-Q_{N_2}) = 0$: CP maps a valid LLC-conserving configuration to another valid LLC-conserving configuration with the same action. CP is therefore an exact symmetry of the strong-force LLC, giving $\theta_{\mathrm{QCD}} = 0$ independently of the instanton argument.

No axion. The Peccei–Quinn axion was introduced specifically to dynamically relax θ_{QCD} to zero. Since RA gives $\theta_{\mathrm{QCD}} = 0$ structurally, no axion field is required. RA predicts the axion does not exist, consistent with decades of non-observation in dedicated searches.

10.2. The black hole information paradox: dissolved by severance

Hawking's calculation shows that black holes emit thermal radiation. If the radiation is exactly thermal, it carries no information about infalling matter. This appears to violate the unitarity of quantum evolution, which requires information to be preserved.

A black hole in RA is a region where the local actualization density approaches the bandwidth ceiling c/ℓ_P, forming a causal severance: a topological boundary in the DAG beyond which no outgoing causal edges exist. This is not the destruction of causal structure but its partition into two causally disconnected components (Paper III).

The LLC holds independently on each component (the Graph Cut Theorem, Lean-verified). The conserved charges $(Q_{N_1}, Q_{N_2}, Q_{N_3/N_4}, p^\mu)$ that flow into the severance from the parent spacetime are encoded in the initial conditions of the daughter subgraph. Information is not destroyed; it is partitioned. The book thrown into the black hole has its causal history encoded in the daughter spacetime's initial LLC conditions, fully accessible to observers inside the daughter but inaccessible to exterior observers, because there are no outgoing causal edges from daughter to parent.

Why Hawking radiation appears thermal. An exterior observer cannot access the daughter subgraph. Tracing over the daughter's degrees of freedom produces a mixed state at the severance boundary—thermal radiation. This is information partition, not information loss. The total system (parent + daughter) evolves unitarily; the exterior subsystem evolves non-unitarily only because it is genuinely causally disconnected from part of the total system.

Proposition 8 (No information loss)**.** *The LLC holds on both sides of every causal severance. Unitarity is preserved for the total system (parent + daughter). The apparent information loss for an exterior observer is a consequence of causal disconnection, not of entropy increase. The black hole information paradox does not arise in RA.*

Prediction: no Page curve. The Page curve predicts that if a black hole radiates into a pure total state, the exterior radiation entropy should first increase and then decrease back to zero as the black hole evaporates. In RA, the interior is genuinely causally disconnected—not a purification partner accessible to exterior observers. The exterior radiation is a genuinely mixed state

throughout evaporation, not the marginal of a pure total state. The exterior entropy therefore increases monotonically throughout evaporation and does not decrease. RA predicts no Page curve, in contrast to holographic AdS/CFT accounts. This is a testable difference.

10.3. The baryon asymmetry as a severance initial condition

The observable universe contains approximately 10^{80} baryons and essentially no antibaryons, with baryon-to-photon ratio $\eta_b = n_B/n_\gamma \approx 6.1 \times 10^{-10}$. Sakharov's conditions for dynamical baryogenesis require: (1) baryon number violation, (2) C and CP violation, (3) departure from thermal equilibrium.

Baryon number is exactly conserved in RA. Sakharov condition (1) is therefore structurally absent. This does not mean RA cannot account for η_b; it means the question is posed in the wrong framework.

The Big Bang in RA is a severance nucleation event: our universe is a daughter spacetime whose initial conditions are encoded in the LLC boundary conditions at the severance boundary (Paper III). Among these initial conditions is a net baryon number $B = N_B \neq 0$, inherited from the parent spacetime via the LLC charge balance at the boundary.

The baryon asymmetry was not dynamically generated during cosmic evolution. It is an initial condition of the daughter spacetime, imprinted at the Planck scale by the LLC at the severance boundary. Baryogenesis as a cosmic process does not occur in RA.

Observational status. If the baryon asymmetry is a severance initial condition, no dynamic baryogenesis process occurs and no signatures of electroweak baryogenesis or leptogenesis should appear in precision CMB data. The distinctive RA prediction is the absence of baryon isocurvature perturbations correlated with any baryogenesis epoch at any scale. Future precision measurements (CMB-S4, Simons Observatory) may provide a discriminating test.

The quantitative value $\eta_b \approx 6.1 \times 10^{-10}$ is in principle derivable from the LLC boundary conditions at the severance—a property of the BDG action at Planck density, calculable via the BDG-filter simulation. This is an open computational target.

11. The Selective Regime and Motif Selection

The BDG filter does not treat all depth profiles equally. In the selective regime ($\mu \approx 3$–5), the filter exhibits a characteristic selection pattern (Monte Carlo, 3×10^5 samples per density):

μ	ΔN_1	ΔN_2	ΔN_3	ΔN_4
1.0	−0.34	+0.32	−0.14	+0.03
2.0	−0.12	+0.58	−0.66	+0.16
4.0	−0.05	+1.10	−2.51	+1.32
4.7	−0.06	+1.17	−3.17	+1.93
7.0	−0.02	+0.60	−2.66	+2.18

Here $\Delta N_k = \mathbb{E}[N_k \mid S > 0] - \mathbb{E}[N_k]$ is the shift induced by the BDG filter.

The pattern is universal across densities:

- N_1 always decreased ($c_1 = -1$ penalizes shallow parents),

- N_3 always decreased ($c_3 = -16$ heavily penalizes depth-3),
- N_2 and N_4 increased ($c_2 = +9$, $c_4 = +8$ reward depth-2 and depth-4).

The filter selects for wide, deep, sparse structure: motifs with rich depth-2 and depth-4 ancestry but few shallow (N_1) and depth-3 (N_3) connections. This is the discrete origin of the observed matter spectrum: stable particles are precisely those motifs whose depth composition passes the BDG filter with high probability.

12. Epistemic Status of Results

Result	Method	Status
Topology and structure		
5 topology types exhaust SM	Lean 4 (124 cases)	LV
Confinement depths $L = 3, 4$	Lean 4	LV
Baryon chirality (13 theorems)	Lean 4	LV
Charge quantization in $e/3$	N_2 winding in 3+1D	DR
Coupling constants		
$\alpha_{\rm EM}^{-1} = 137$ (integer core)	Lean 4 (`norm_num`)	LV
$\alpha_{\rm EM}^{-1} = 137.036$ (full value)	Discrete Dyson equation	CV
$\alpha_s(m_Z) = 1/\sqrt{72}$	BDG UV fixed point	CV
$\alpha_s^{\rm IR} = 1/3$	BDG RG IR fixed point	CV
Koide $K = 2/3$ (leptons)	Lean 4	LV
Koide breaking K_q, K_ℓ pattern	Casimir formula identification	PI (3–8%)
Cabibbo angle $\lvert V_{us}\rvert = \sin(2/9)$	$\mathrm{SU}(3)_{\rm gen}$ motivated identification	PI (1.8%)
$\lvert V_{cb}\rvert = (2/\pi)\delta = 0.0424$	Angle-displacement identification	PI (0.4%)
Neutrinos structurally Majorana-compatible	Sign flip in $\sqrt{m_k}$	PI
$\sum m_\nu \approx 59\,\mathrm{meV}$	Koide + oscillation Δm^2	CV
Mass cascade		
$\mu_p = \alpha_{\rm EM}/32$ (self-consistency)	Requires $N_{\rm eff} = L_q^3 = 64$	CN
$m_p = m_P \alpha_{\rm EM}^5/2^{28} = 941\,\mathrm{MeV}$	μ^5 cascade identification	PI (0.3%)
$m_n = m_p$ (leading order)	Same BDG topology	DR
$m_H = 133 m_p = 125.2\,\mathrm{GeV}$	Depth-2 + EM ratio identification	PI (0.06%)
$r_p = L_q \ell_C = 0.84\,\mathrm{fm}$	Confinement radius identification	PI (0.03%)
$f_0 = 17.32 \times \alpha_s(2m_p) = 5.40$	BDG enumeration + ext. SM RG	CV (0.3%, w/ ext. input)
$m_\pi \approx 124\,\mathrm{MeV}$	$(2/3)^5 m_p$ numerical observation	PI (11%)
$m_W = 6^{5/2} m_p = 83\,\mathrm{GeV}$	Handedness sector identification	CN (3.3%)
$\mu_{\rm QCD} = 4.7118$	$\exp(\sqrt{4\Delta S^*})$, canonical D4U02	CV (0.03%)
Force ranges		
EM $1/r^2$	Causal 2-sphere geometry	DR
Weak $R_W = \hbar/(m_W c)$	BDG debt lifetime (unified principle)	DR
Strong $V(r) = \sigma r$	LLC flux tube (unified principle)	DR
Regge $m^2 \propto J$	Flux tube $r \propto \sqrt{J}$	DR
Gauge structure		
$\mathrm{SU}(3)_{\rm col} \times \mathrm{SU}(2) \times \mathrm{U}(1)$	Sign mechanism + Berry identification	PI
$\mathrm{SU}(3)_{\rm col} \times \mathrm{SU}(3)_{\rm gen}$ GUT	$\mu \to \infty$ structural conjecture	CN
$\sin^2\theta_W = 3/8$ at GUT scale	SU(5)-type embedding (if GUT holds)	DR (conditional)
Structural predictions / absences		
Absence of SUSY partners	BDG algebra not generated	DR
Absence of WIMPs	Paper III ($P_{\rm act}$ filter)	DR
Exact proton stability	LLC applied to N_2 winding	LV
$\theta_{\rm QCD} = 0$ exactly	DAG has no instantons	DR
Absence of axion	$\theta_{\rm QCD} = 0$ structurally	DR
No Page curve	Severance partition (not purification)	DR
Baryon asymmetry as initial condition	Severance boundary LLC	DR

Legend. LV = Lean-verified theorem (machine-checked at the stated layer; BDG coefficient uniqueness is LV at the arithmetic-inversion layer and supplemented by Yeats (2025) at the geometric-

identification layer, see Paper I §3.4). DR = derived (all steps explicit within RA). CV = computation-verified (exact derivation plus numerical evaluation). PI = phenomenological interpolation (motivated identification supported by numerical evidence, not yet derived from first principles). CN = conjecture (stated path to closure, no current derivation).

Reading the table. A PI label does not mean "wrong"—several PI results in this table match observation to a part in 10^3 or better—but it does mean that the agreement is not yet a derivation. Sharpening a PI to a DR or CV is the natural target for future work on the relevant section.

13. What the Standard Model Gets Right, and Why

The Standard Model is not wrong. It is a spectacularly accurate effective description of the BDG filter's output in the selective regime. Its symmetry groups encode the geometric structure of the BDG depth profile (at the DR/PI level of this paper). Its coupling constants are identified with eigenvalues or fixed points of the BDG integer arithmetic. Its particle spectrum corresponds, at the topology-class level, to the five stable renewal motifs in $d = 4$ (Lean-verified).

What the Standard Model cannot explain—why these groups, why these constants, why these particles—is not a failure of the Standard Model but, on this picture, a consequence of the fact that it describes the output without seeing the input. The input, in RA's framing, is the BDG action; the output is the Standard Model. The extent to which the RA programme has actually *derived* the Standard-Model content from BDG primitives is what the epistemic-status table in §12 attempts to characterize honestly.

14. Conclusion

From the BDG integers $(1, -1, 9, -16, 8)$ and the sequential growth rule of Paper I, the present paper develops the matter-and-force programme of Relational Actualism. Its strongest claims are structural: the five topology types, the renewal architecture, the confinement depths, and the organization of force mediation by discrete graph structure. On top of that structural core, the paper presents a number of numerical evaluations and phenomenological identifications, including candidate relations for couplings, masses, flavour parameters, and force ranges. Some of these are strikingly close to observed values, but they do not all stand at the same epistemic level, and should not be read as if they do.

The paper therefore does not claim complete closure of the particle sector. Rather, it establishes a disciplined programme in which a substantial part of particle structure is constrained by discrete topology and renewal, while several quantitative bridges to observation remain to be sharpened, proved, or replaced. The particles are not, in this picture, excitations of fields on a background manifold. They are finite local topology classes—renewal motifs—that reproduce themselves under the BDG-filtered growth dynamics.

Open targets. The following are the most important unresolved items in this paper's programme: (i) full derivation of the Higgs-sector mass relation from BDG dynamics; (ii) closure of the W-mass conjecture; (iii) first-principles derivation of N_{eff} in the proton mass cascade; (iv) Haar-measure or

algebraic derivation of the Cabibbo angle and CKM parameters; (v) closure of the gauge-group identification at theorem level from BDG primitives to the continuum Standard-Model group; (vi) first-principles derivation of the $\alpha_s(\mu)$ RG profile between the BDG UV and IR fixed points (which would close the f_0 bridge); (vii) experimental discriminants separating RA predictions from Standard Model expectations; (viii) an RA-native mechanism for the pion mass residual (see §4.6), whether via a discrete Goldstone-like structure rooted in DAG directionality or via a structurally distinct RA-native mechanism with no Standard Model analog.

Paper III develops the macroscopic consequences: gravity, cosmology, and causal severance. Paper IV develops the emergence of biological complexity from the same underlying percolation threshold.

Acknowledgments

As in Paper I, this work was developed in collaboration with Claude (Anthropic) for computation and manuscript preparation, and with ChatGPT (OpenAI) for external review and proof structure. The ontological commitments, physical reasoning, and framework direction are the author's.

References

[1] D. M. T. Benincasa and F. Dowker, "The scalar curvature of a causal set," *Phys. Rev. Lett.* **104**, 181301 (2010).

[2] F. Dowker and L. Glaser, "Causal set d'Alembertians for various dimensions," *Class. Quantum Grav.* **30**, 195016 (2013).

[3] K. Yeats, "Hockey-stick-style identities and the moments of causal intervals in d-dimensional Lorentzian causal diamonds," *Class. Quantum Grav.* **42**, 145003 (2025).

[4] R. L. Workman et al. (Particle Data Group), "Review of Particle Physics," *PTEP* **2022**, 083C01 (2022), and 2024 update.

[5] N. Aghanim et al. (Planck Collaboration), "Planck 2018 results. VI. Cosmological parameters," *Astron. Astrophys.* **641**, A6 (2020).

[6] Y. Koide, "New viewpoint in the charged lepton mass relation," *Phys. Lett. B* **120**, 161 (1983).

[7] A. Wyler, "L'espace symétrique du groupe des équations de Maxwell," *C. R. Acad. Sci. Paris* **272A**, 186 (1971).

[8] N. Cabibbo, "Unitary symmetry and leptonic decays," *Phys. Rev. Lett.* **10**, 531 (1963); M. Kobayashi and T. Maskawa, "CP-violation in the renormalizable theory of weak interaction," *Prog. Theor. Phys.* **49**, 652 (1973).

[9] H. Georgi and S. L. Glashow, "Unity of all elementary-particle forces," *Phys. Rev. Lett.* **32**, 438 (1974).

[10] R. D. Peccei and H. R. Quinn, "CP conservation in the presence of pseudoparticles," *Phys. Rev. Lett.* **38**, 1440 (1977).

[11] D. N. Page, "Information in black hole radiation," *Phys. Rev. Lett.* **71**, 3743 (1993).